

BG4104 Group Project

Supervised and Unsupervised Learning for Gastroesophageal Reflux pH Time Series

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Abstract—This study asks whether acid and non-acid source labels in oesophageal pH time series are only predictable with supervision or also emerge as natural structure. We reuse the audited 120-s window pipeline and supervised findings reported previously, then focus on label-free K-Means and Gaussian mixture models (GMM), principal component analysis (PCA), independent component analysis (ICA), Gaussian random projection (RP), and all six reduction–clustering combinations. The unsupervised fit never uses labels; labels return only for post-hoc adjusted Rand index (ARI), normalised mutual information (NMI), and matched accuracy. On the locked supervised test set, polynomial SVM reached 0.8438 accuracy and 0.8525 F1. On all nine standardised Part-B features, two-component full-covariance GMM was the strongest raw clustering (ARI 0.1233, matched accuracy 0.6759), although label-free criteria preferred six clusters. Four PCs retained 95.81% variance. Of the six required combinations, PCA(4D)+K-Means led by ARI (0.0685), only 0.0015 above raw K-Means; reduction substantially harmed GMM. Thus, natural pH geometry exists, but it describes multiple level, variability, and extreme-event states rather than reconstructing the two record-level labels. Supervision is preferable for the stated classification target, while unsupervised learning remains useful for subtype discovery, quality control, drift monitoring, and annotation support.

Index Terms—gastroesophageal reflux disease, pH time series, clustering, Gaussian mixture model, PCA, ICA, random projection

I. INTRODUCTION

Gastroesophageal reflux disease (GERD) is clinically evaluated using symptoms and, where required, objective tests such as ambulatory reflux monitoring. Oesophageal acid exposure time below pH 4 is informative, but a low reading alone is not a complete reflux diagnosis; modern consensus requires the measurement to be interpreted with other evidence [1], [2]. This distinction matters here because the supplied labels describe the source recording, not second-by-second clinical events. GERD is also increasingly relevant in Asia, where non-erosive phenotypes are common and automated analysis remains underused [3], [4].

The same data support two related but different machine-learning questions. Supervised learning asks how accurately labelled windows can be classified. Unsupervised learning removes those labels and asks whether pH level, variability, extrema, and temporal trend naturally reproduce the same partition. The latter is scientifically useful even when it fails: disagreement can reveal that the binary labels compress a richer continuum of physiological or acquisition states.

This report therefore addresses four questions: (1) do K-Means and covariance-aware GMM recover two label-aligned clusters on all features; (2) what structure do PCA, ICA, and RP preserve; (3) which of six reduction–clustering combinations best recovers the known partition; and (4) when is supervised or unsupervised analysis preferable in deployment? Data auditing, exploratory analysis, window selection, feature engineering, and the full neural-network/SVM/ k -NN study have already been documented [5]; they are summarised only where needed for a fair cross-part interpretation. The contribution here is the complete three-stage unsupervised analysis, including pre-specified hypotheses, label-free selection, negative results, and explicit limitations.

II. METHODOLOGY

A. Dataset, EDA, and Preprocessing

The parser accepted only two-column timestamp/pH rows and audited malformed trailing content. It found 172,597 valid readings, no missing values, duplicate timestamps, or pH values outside 0–14. The filenames conflict with the exact class counts in the brief, so the auditable count mapping was retained: 86,400 rows as Acid and 86,197 as Non-acid. The median timestamp increment is 1,000 ms, not 1 ms. Consequently, a 120-sample non-overlapping window represents approximately 120 s. Development-only cross-validation previously selected this window from 15, 30, 60, 120, and 300 samples [5]. It yields 1,438 windows (720 Acid, 718 Non-acid).

Part A used 22 features: nine required summaries plus 13 quantile, difference, autocorrelation, skewness, and kurtosis features. The revised Part-B specification explicitly defines a nine-dimensional input, so Part B uses mean, standard deviation, minimum, maximum, range, median, slope, $\text{pH} > 7$ ratio, and $\text{pH} < 6$ ratio. StandardScaler was fitted without labels. Labels were excluded from scaling, transformation, model selection, and fitting, and were used only after fitting for ARI, NMI, matched cluster accuracy, and marker shapes. This prevents circular selection while retaining a post-hoc test of label recovery.

B. Supervised Learning Summary

The prior study used a fixed stratified 80/20 development/test split; scaling and all selection occurred inside five-fold development cross-validation [5]. It compared feed-forward neural networks, RBF and polynomial SVMs, and k -NN, then evaluated the frozen winners once on 288 test windows. Because those methods, learning/model-complexity curves, uncertainty analyses, and preprocessing decisions are already reported, they are not repeated. The present report imports only the final benchmark needed for Section IV.

C. Unsupervised Design and Hypotheses

Six hypotheses were recorded before interpreting labels: H1, the nine summaries contain a recoverable two-cluster label structure; H2, GMM should beat Euclidean K-Means because correlated features imply anisotropic clusters; H3, at most three PCs retain at least 90% variance; H4, both components in 2D ICA are clearly non-Gaussian; H5, PCA at the 95% threshold improves or preserves ARI; and H6, 2D RP is reasonably stable over five seeds although the $\epsilon = 0.1$ Johnson–Lindenstrauss (JL) bound is too conservative for nine original dimensions. H1–H2 concern Stage 1, H3–H4–H6 Stage 2, and H5 Stage 3.

D. Stage 1: Clustering on Nine Features

K-Means minimises within-cluster squared Euclidean distance [6]. We tested $k = 2, \dots, 6$ with both k-means++ and random initialisation, using inertia, Silhouette [7], and Davies–Bouldin (DB) [8]. Euclidean geometry is defensible after standardisation, but correlated derived features make its spherical, equal-scale partition restrictive.

GMM fitted $n = 2, \dots, 6$ components by expectation maximisation [9] with full, tied, diagonal, and spherical covariance. Its log-likelihood implicitly measures covariance-scaled (Mahalanobis) separation, allowing tilted ellipses. BIC and AIC evaluated fit–complexity trade-offs; Silhouette, DB, convergence, and runtime provided complementary diagnostics. The $n = 2$ setting was selected by minimum BIC without labels. ARI [10], NMI, and matched accuracy were computed only afterwards. Global label-free optima were also retained rather than forcing the binary answer.

E. Stage 2: Dimensionality Reduction

PCA computed all nine orthogonal components [11]; cumulative variance determined the minimum dimensions for 90%, 95%, and 99%, while loadings supported feature-level interpretation. ICA [12] was run at 2, 3, 5, 7, and 9 components. Convergence iterations, reconstruction error, and excess kurtosis measured optimisation and non-Gaussianity. Physiological labels for independent components were treated as exploratory, not causal claims.

ICA was also evaluated as a dimensionality reduction technique, but its effectiveness was limited because ICA assumes that the observed variables are linear mixtures of statistically independent latent sources. In contrast, this dataset consists of summary statistics computed from a single concatenated pH trace. Features such as median_ph are deterministic functions of the same samples used to compute mean_ph, resulting in very high correlations ($r \approx 0.99$). Consequently, these features do not satisfy ICA’s assumption that the observed variables are linear mixtures of independent sources, making ICA unsuitable for this feature representation.

Gaussian RP was tested at 2, 3, 5, 7, and 9 dimensions with seeds 100–104. On 200 sampled windows, we compared pairwise distances by correlation, relative error, and the fraction within 10% error. The JL sufficient bound was calculated for $n = 1,438$ and $\epsilon = 0.1$ [13]. K-Means and GMM were then run after every trial; mean and standard deviation quantified downstream stability.

F. Stage 3 and Reproducibility

Both clusterers were applied to PCA(4D), ICA(2D), and RP(2D), giving six required combinations. PCA dimension was selected by the label-free 95% variance rule; ICA and RP used the required 2D representation. The combination best recovering the known binary partition was ranked post hoc by ARI, with NMI and Silhouette as supporting metrics. All Part-B random seeds were fixed at 100 (RP trials 100–104). Python, NumPy, pandas, SciPy, scikit-learn, Matplotlib, source scripts, configuration, CSV outputs, and a Linux README are included for reproduction [14].

G. Evaluation Discipline

Model selection and interpretation were deliberately separated. Silhouette, DB, BIC, AIC, convergence, and stability were available during unsupervised selection; ARI, NMI, matched accuracy, and true-label marker shapes were revealed only after fitting. Raw and reduced clusterers used the same 1,438 windows, whereas the supervised benchmark used a locked holdout and more features. Consequently, cross-paradigm accuracy is reported only as an orientation, while within-Part-B ARI differences support the reduction comparisons.

The metrics answer different questions. Silhouette and DB evaluate compactness and separation in the representation supplied to the clusterer; BIC and AIC trade GMM likelihood against parameter count. None targets the source labels. By contrast, ARI and NMI are permutation-invariant external

TABLE I
STAGE-1 TWO-CLUSTER RESULTS

Method	Sil.	DB	ARI	NMI	Match
K-Means, $k = 2$	.412	1.172	.067	.054	.630
GMM-full, $n = 2$	.250	2.114	.123	.110	.676

audits, while matched accuracy requires an optimal post-hoc label mapping. A method may therefore have a high Silhouette but low ARI without contradiction: it has found a clean partition that is different from the supplied binary taxonomy.

Robustness was assessed at the level supported by the available data. K-Means used two initialisation schemes, every GMM reported convergence and covariance sensitivity, and RP used five independent seeds. Exact configurations and per-trial outputs were retained instead of reporting only the winning run. These checks quantify optimisation and representation stability, but they do not replace patient-level confidence intervals because the dataset contains only two long source recordings.

III. RESULTS

A. Supervised Benchmark

Fig. 1 condenses the previously reported supervised results. Polynomial SVM gave the highest locked-test accuracy/F1/recall (0.8438/0.8525/0.9028), whereas RBF-SVM gave the highest ROC-AUC (0.9217). Bootstrap intervals overlapped and paired McNemar tests found no significant winner; the benchmark is therefore a useful point estimate, not patient-level validation [5].

B. Stage 1: Raw-Feature Clustering

Fig. 2 shows that K-Means Silhouette rises through $k = 6$ and GMM BIC is lowest for six full-covariance components. Thus the global label-free optimum is not the requested binary partition. At $k = n = 2$, Table I shows a metric tension: K-Means has the higher Silhouette (0.4116), but full-GMM has 1.84 times the ARI and higher NMI/matched accuracy. Silhouette favours compact Euclidean clusters; it need not reward label-aligned anisotropic components.

The two K-Means initialisations converged to the same $k = 2$ solution in 6–7 iterations. Full-GMM converged in 19 EM iterations. Within $n = 2$, full covariance had BIC $-5,746$ and ARI 0.1233, whereas spherical covariance had BIC 31,747 and ARI 0.0446. This large loss when ellipses are forced into spheres supports H2. Fig. 3 makes the substantive result visible: marker shapes (true labels) remain mixed inside both predicted colours. H1 is therefore rejected; natural structure exists, but it is not primarily the binary source label.

C. Dimensionality Reduction (Stage 2)

1) *PCA*: PCA required four components to cross both the 90% and 95% variance thresholds: PC1, PC1–2, PC1–3, and PC1–4 retained 49.49%, 77.49%, 88.43%, and 95.81%, respectively, with six components reaching 99.41%. The hypothesised two-to-three components are therefore insufficient,

and H3 is rejected. This holds beyond w120: confirmed by a window sweep (w15–w300, Fig. 5), three components never exceed 88.7% at any window size tested, and four are required for 90% throughout.

In Fig. 4, PC1 is dominated by mean (0.466), median (0.453), and below-6 ratio (-0.389), an acid-exposure or level axis. PC2 is dominated by range (0.607), standard deviation (0.594), and minimum (-0.341), a variability axis capturing low-pH extremes. The loadings are coherent summaries of the underlying signal, but coherence alone does not establish distinct physiological mechanisms.

2) *ICA*: All ICA configurations converged. The two-component solution converged in two iterations with component excess kurtoses of -0.07 and 6.90 : only one recovered direction is strongly non-Gaussian. This rejects H4, which assumed two independent non-Gaussian sources. Raising the target dimension increased optimisation effort and yielded more extreme kurtosis, but a numerically independent direction is not necessarily an independent physiological process; ICA’s mixing assumption is weak here, since the nine features are deterministic functions of a single pH signal rather than a linear mixture of separate sources.

3) *Random Projection*: At two dimensions, random projection preserved pairwise distances poorly: distance correlation was 0.746 ± 0.082 , mean relative error 0.483 ± 0.096 , and only $9.0\% \pm 5.2\%$ of distances fell within 10% of their original values. Downstream K-Means on the projection gave an ARI of 0.071 ± 0.068 across five seeds. The Johnson–Lindenstrauss sufficient bound for $\epsilon = 0.1$ was 6,232 dimensions—far above the nine available—so JL provides no usable compression target at this dimensionality. This reflects the asymptotic nature of the bound in the sample count, not a failure of RP as a method.

The bound exceeds the nine available dimensions at every window (5,446–8,014, Fig. 7), confirming that the limitation is structural rather than specific to w120. As Fig. 6 shows, additional projection dimensions improve distance correlation but do not monotonically improve label agreement.

D. Stage 3: Six Combined Experiments

Table II and Fig. 8 report all six combinations. PCA(4D)+K-Means is the requested winner by ARI (0.0685), NMI 0.0546, Silhouette 0.4197, and matched accuracy 0.6314. Its ARI gain over raw K-Means is only 0.0015. PCA+GMM loses 0.0693 ARI versus raw GMM, while ICA harms both clusterers. RP produces the highest Silhouette (0.6704) and NMI (0.1735) for K-Means, yet ARI remains 0.0638. Compact geometry therefore does not imply recovery of the clinical partition.

IV. DISCUSSION

A. Closing the Hypothesis Loop

Only H2 and the qualified H6 were supported. H1 failed because even full-GMM ARI was only 0.1233 and label-free selection preferred six states. H3 failed because three PCs retained 88.43%, not 90%. H4 failed because one 2D ICA component was nearly Gaussian. H5 failed: PCA barely

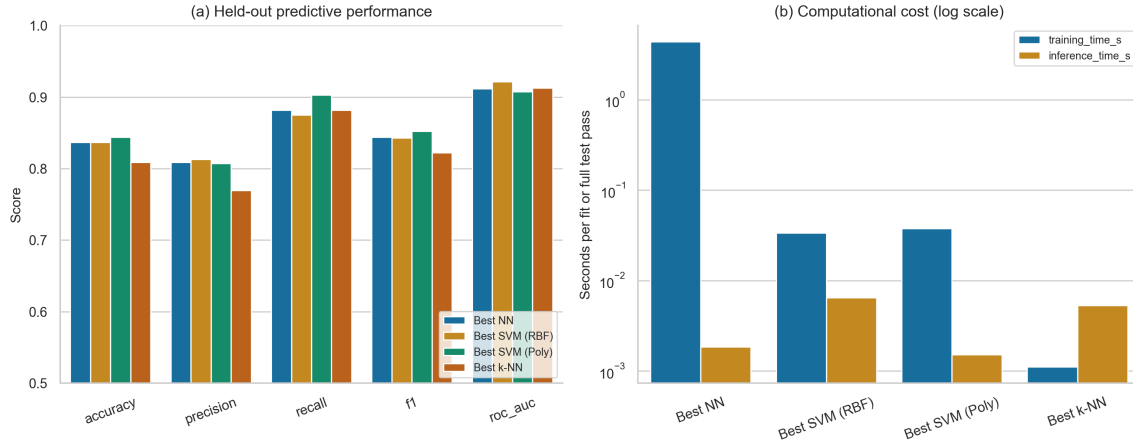


Fig. 1. Supervised benchmark reproduced from the prior report: held-out predictive metrics (left) and computational cost (right).

TABLE II
STAGE-3 REDUCTION AND CLUSTERING RESULTS (RANKED BY ARI)

Rank	Reduction	Clusterer	Dim.	ARI	NMI	Silhouette	Match	Δ ARI vs. raw
1	PCA	K-Means	4	.0685	.0546	.4197	.6314	+.0015
2	RP	K-Means	2	.0638	.1735	.6704	.6266	-.0032
3	RP	GMM	2	.0631	.1636	.6649	.6259	-.0602
4	PCA	GMM	4	.0540	.0419	.4048	.6168	-.0693
5	ICA	GMM	2	.0403	.0728	.4946	.6008	-.0830
6	ICA	K-Means	2	.0235	.0557	.5410	.5772	-.0435

changed K-Means and substantially harmed GMM. H6 was supported in the intended qualified sense: five-seed 2D results were usable but variable, while the 6,232-dimensional JL bound was too conservative to guide a nine-feature problem. Keeping these negative outcomes avoids retrospectively tuning to the labels.

B. What the Algorithms Reveal

K-Means and GMM optimise different notions of structure. K-Means rewards compact Euclidean cells and consequently obtains the best two-cluster Silhouette. Full-GMM can model correlated, stretched components and better aligns with the labels despite a worse Euclidean Silhouette. The covariance audit is especially diagnostic: spherical GMM loses most of full-GMM’s ARI. Nevertheless, both label-free criteria favour

six clusters, suggesting windows are organised by several pH-level, variability, and extreme-event regimes. A binary record label compresses those regimes and need not coincide with density modes.

Reduction also creates a trade-off. PCA preserves global variance efficiently and makes the first two axes interpretable, but high variance is not the same as cluster-discriminative information. Discarded low-variance directions may matter to GMM, explaining why PCA harmed it. ICA searches for statistical independence rather than class separation; the weakly non-Gaussian first component and poor downstream ARI show that independent mathematical sources need not map to Acid/Non-acid. RP can create compact 2D groups and high Silhouette while distorting enough pairwise relationships

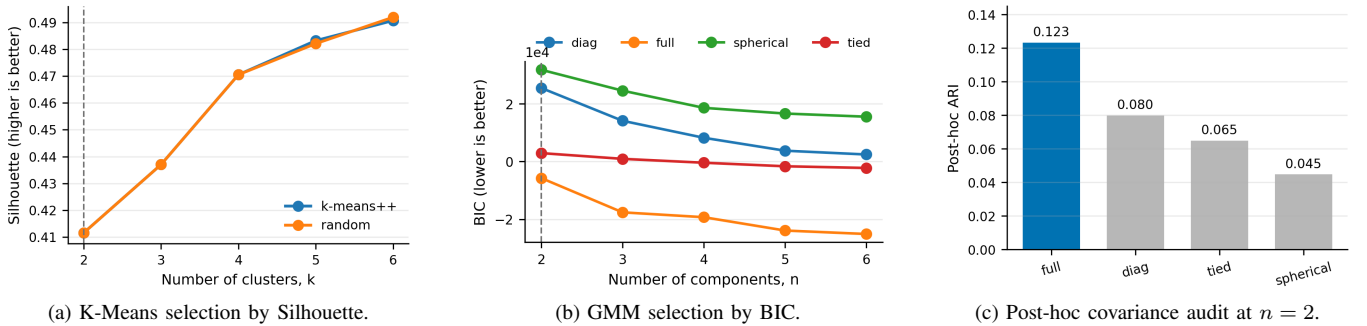


Fig. 2. Stage-1 model-selection diagnostics. Dashed lines mark the required two-cluster comparison. Labels were not used to select k , n , or covariance; panel (c) is a post-hoc audit.

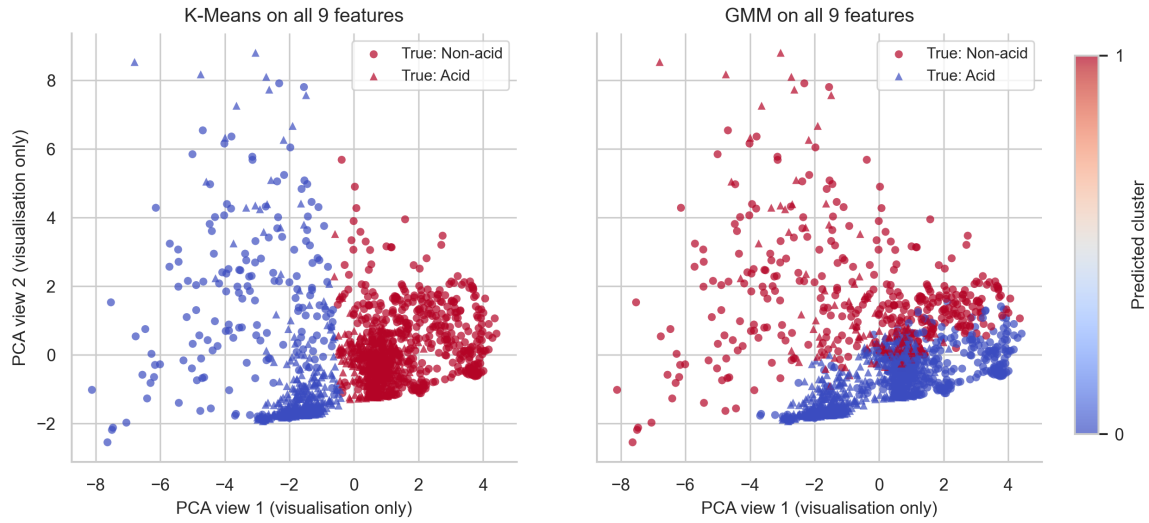


Fig. 3. Two-cluster assignments on all nine features, displayed in a PCA view used only for visualisation. Colour is predicted cluster; marker shape is the post-hoc true label.

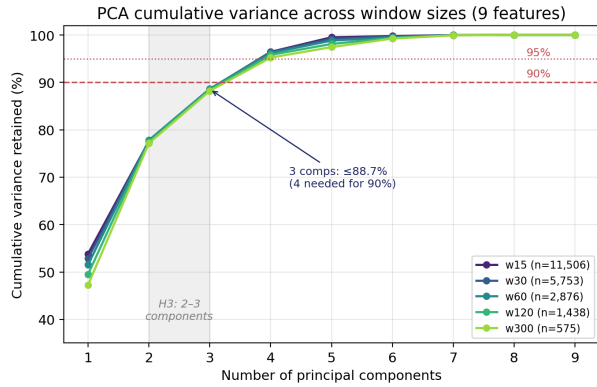
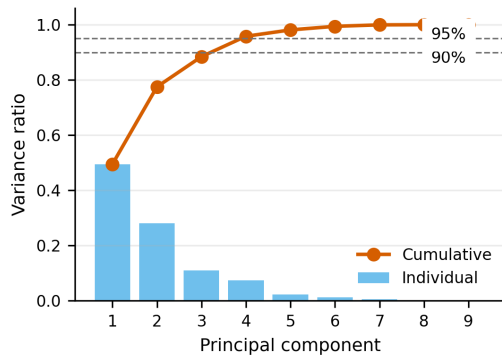
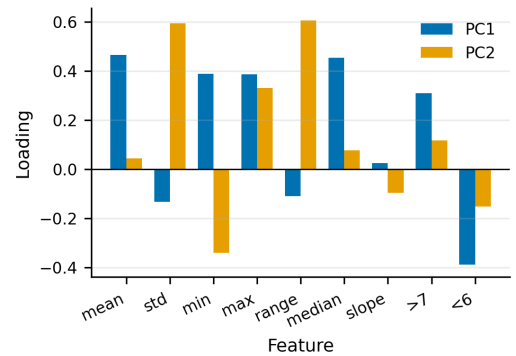


Fig. 5. PCA cumulative variance across window sizes. Three components never exceed 88.7%; four are required to reach 90% at every window, rejecting H3 independently of window choice.

to misalign labels. No reduction method helps both clusterers consistently, so H5's denoising intuition is not general.



(a) Explained and cumulative variance.



(b) Loadings of the first two components.

Fig. 4. PCA diagnostics. Four components, not the hypothesised three, are needed to cross both the 90% and 95% thresholds.

C. Supervised Versus Unsupervised Learning

Fig. 8(c) gives a deliberately qualified comparison. Polynomial SVM accuracy 0.8438 comes from a locked 20% test set and 22 features. PCA+K-Means matched accuracy 0.6314 and raw full-GMM 0.6759 are transductive post-hoc scores after fitting all 1,438 windows with nine features. These are not identical tests, so the 0.21 gap cannot be attributed only to supervision. Even with that caveat, labels clearly provide information needed to reproduce the stated target. If prediction of the supplied Acid/Non-acid definition is the goal, supervised SVM is preferable; use RBF-SVM when ranking and threshold adjustment matter more than default-threshold F1.

Unsupervised analysis remains preferable when expert labels are absent or costly, when the task is discovery rather than replication of an existing taxonomy, or when monitoring unlabelled deployment data for drift and sensor anomalies. The six-cluster result may generate candidate subtypes or representative cases for annotation. Its advantage is scalable exploration; its risk is assigning clinical meaning to geometric

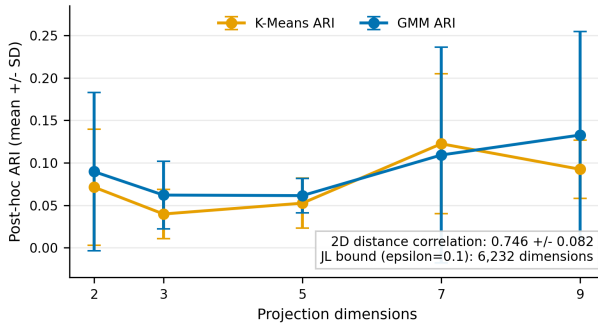


Fig. 6. Five-seed random-projection clustering stability across projection dimensions. Error bars denote one standard deviation. Increasing the projection dimension generally improves stability, but label agreement remains non-monotonic.

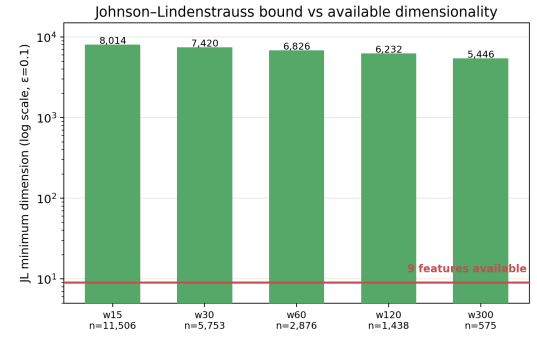


Fig. 7. Johnson-Lindenstrauss minimum dimension ($\epsilon = 0.1$) across window sizes, compared with the nine available features. The theoretical bound exceeds the original dimensionality at every tested window size.

groups without independent validation.

D. Limitations and Clinical Implications

The dataset contains two long class recordings, not many independent patients. Non-overlapping windows reduce direct overlap but remain temporally related; neither supervised holdout nor transductive clustering demonstrates new-patient generalisation. The filename/count conflict, 1-ms/1,000-ms discrepancy, and higher $\text{pH} < 4$ fraction in the nominal Non-acid recording further constrain interpretation. Part A and Part B also use different feature counts and evaluation scopes. ARI/NMI are legitimate post-hoc audits but cannot be used to choose unsupervised hyperparameters without circularity.

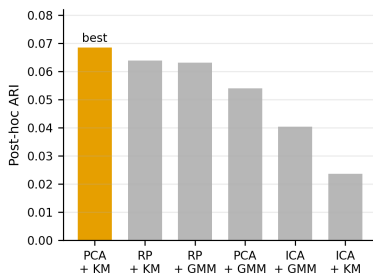
Future work should collect multi-patient, multi-centre pH-impedance data and split evaluation by patient. Symptom timing, medication, impedance, endoscopy, and demographic covariates could distinguish acid exposure from source-recording artefacts. Cluster stability should then be checked across sites and time, followed by blinded clinician review. Until then, these models are tools for screening, prioritisation, data quality, and hypothesis generation, not autonomous GERD diagnosis.

V. CONCLUSION

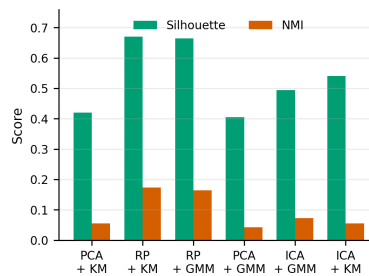
This report integrates a previously validated supervised pipeline with a complete three-stage unsupervised study. Full-covariance GMM is the best raw two-cluster model, confirming that anisotropic geometry matters, but its ARI 0.1233 and the label-free preference for six clusters reject a simple two-group account. PCA needs four components for 95.81% variance; ICA provides only one strongly non-Gaussian 2D source; and 2D RP is moderately variable and does not preserve distances within 10%. PCA(4D)+K-Means leads the six required combinations, but its improvement over raw K-Means is negligible and no reduction helps both clusterers. Supervision therefore remains the appropriate choice for the supplied binary target, whereas unsupervised methods add value by exposing richer structure, supporting annotation, and monitoring data. The central methodological result is that geometric compactness, variance preservation, and agreement with clinical labels answer different questions and must be reported separately.

VI. ACKNOWLEDGMENT

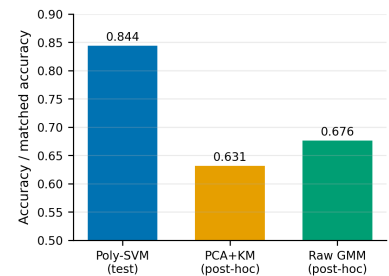
The authors thank the course instructor for providing the data and project framework. OpenAI Codex assisted with code drafting, result verification, and language polishing; every



(a) Post-hoc label recovery measured by ARI.



(b) Geometric compactness versus label agreement.



(c) Qualified supervised-unsupervised comparison.

Fig. 8. Stage-3 and cross-paradigm results. Panel (a) compares post-hoc recovery of the supplied binary labels. Panel (b) shows that geometric compactness and label agreement can select different winners. Panel (c) provides only an orientation because supervised accuracy and post-hoc matched accuracy use different feature sets and evaluation scopes.

numerical result and figure was generated from the Python code and saved outputs in the submission directory.

REFERENCES

- [1] C. P. Gyawali, P. J. Kahrilas, E. Savarino *et al.*, “Modern diagnosis of GERD: The Lyon consensus,” *Gut*, vol. 67, no. 7, pp. 1351–1362, 2018.
- [2] C. P. Gyawali, R. Yadlapati, R. Fass *et al.*, “Updates to the modern diagnosis of GERD: Lyon consensus 2.0,” *Gut*, vol. 73, no. 2, pp. 361–371, 2024.
- [3] H.-K. Jung, “Epidemiology of gastroesophageal reflux disease in asia: A systematic review,” *Journal of Neurogastroenterology and Motility*, vol. 17, no. 1, pp. 14–27, 2011.
- [4] K.-L. Goh, Y.-Y. Lee, S. Leelakusolvong *et al.*, “Consensus statements and recommendations on the management of mild-to-moderate gastroesophageal reflux disease in the southeast asian region,” *JGH Open*, vol. 5, no. 8, pp. 855–863, 2021.
- [5] Y. Hu, “Supervised learning assignment: Classification of gastroesophageal reflux ph time series,” 2026, bG4104 supervised learning assignment report, Nanyang Technological University.
- [6] J. MacQueen, “Some methods for classification and analysis of multivariate observations,” *Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability*, vol. 1, pp. 281–297, 1967.
- [7] P. J. Rousseeuw, “Silhouettes: A graphical aid to the interpretation and validation of cluster analysis,” *Journal of Computational and Applied Mathematics*, vol. 20, pp. 53–65, 1987.
- [8] D. L. Davies and D. W. Bouldin, “A cluster separation measure,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. PAMI-1, no. 2, pp. 224–227, 1979.
- [9] A. P. Dempster, N. M. Laird, and D. B. Rubin, “Maximum likelihood from incomplete data via the EM algorithm,” *Journal of the Royal Statistical Society: Series B*, vol. 39, no. 1, pp. 1–38, 1977.
- [10] L. Hubert and P. Arabie, “Comparing partitions,” *Journal of Classification*, vol. 2, pp. 193–218, 1985.
- [11] I. T. Jolliffe, *Principal Component Analysis*, 2nd ed. New York: Springer, 2002.
- [12] A. Hyvärinen and E. Oja, “Independent component analysis: Algorithms and applications,” *Neural Networks*, vol. 13, no. 4–5, pp. 411–430, 2000.
- [13] S. Dasgupta and A. Gupta, “An elementary proof of a theorem of Johnson and Lindenstrauss,” *Random Structures and Algorithms*, vol. 22, no. 1, pp. 60–65, 2003.
- [14] F. Pedregosa, G. Varoquaux, A. Gramfort *et al.*, “Scikit-learn: Machine learning in python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.